

Weather, Calendar, and Neighborhood Context for Accurate and Explainable Urban Mobility Prediction

THE PROBLEM



1. Demand varies by hour, weather, and events



2. Bike shortages and dock overflow reduce service quality

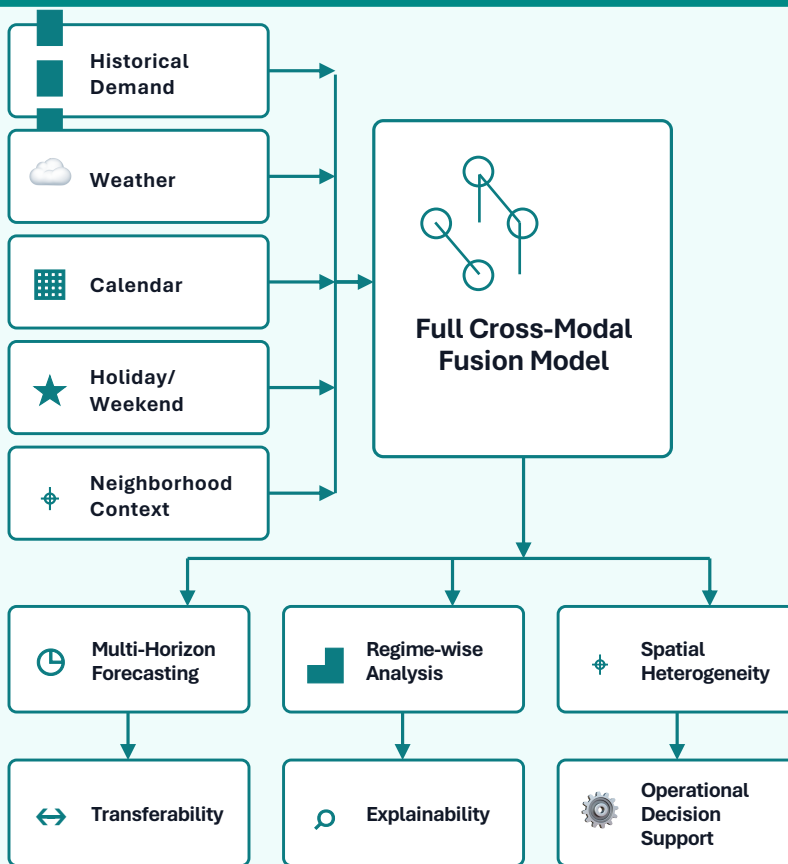


3. Temporal-only forecasting misses urban context



4. Operators need reliable short-term demand prediction

OUR APPROACH



KEY FINDINGS



1. Cross-modal fusion improves forecasting accuracy



2. Best performance varies by demand regime



3. Neighborhood context captures spatial heterogeneity



4. Explainable AI reveals influential factors



5. Forecasts support rebalancing and service planning



Context-aware forecasting enables more reliable, interpretable, and operationally useful shared mobility systems.